

Analysis First-Year Exam, August 2024, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. This problem has two parts.

(i) The function $f : [-1, 1] \rightarrow \mathbb{R}$ is Riemann integrable and $F : [-1, 1] \rightarrow \mathbb{R}$ is defined by the rule

$$F(t) = \int_0^t f(u) \, du.$$

Prove that F is continuous at 0.

(ii) Does the same conclusion follow if f is instead assumed to be Lebesgue integrable? Explain.

2. Let $A = \{a_n : n \in \mathbb{N}\}$ be a subset of the metric space X . For each $n \in \mathbb{N}$ let $f_n : X \rightarrow \mathbb{R}$ be continuous except perhaps at a_n .

(i) Prove that if $f_n \rightarrow f$ uniformly on X then the set of discontinuities of f is finite or countably infinite.

(ii) Show by example that if $f_n \rightarrow f$ pointwise on X then f can have uncountably many discontinuities.

3. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous; assume that

$$\int_0^1 f(t) t^n \, dt = 0$$

for each integer $n > 1$. Prove that the function f is constantly zero.

4. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space that is complete in the sense that each subset of every μ -null set lies in $\mathcal{A}$. Let $f : \Omega \rightarrow \mathbb{R}$ be measurable and let $A \subseteq \Omega$ be μ -null. Prove that if $g : \Omega \rightarrow \mathbb{R}$ is defined by changing the values of f at points of A , then g is measurable.
5. On the measure space Ω , let $(f_n : n \in \mathbb{N})$ be a sequence of μ -integrable functions that converges uniformly to the function f . Prove that if $\mu(\Omega)$ is finite then f is μ -integrable, and show by example that f can fail to be μ -integrable if $\mu(\Omega)$ is infinite.